SQL Code

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-- AIRLINE RESERVATION SYSTEM : Database Creation and Table Implementation
-- Student : R Monisha Bangera (2511021240053)   Course: DBMS, BCA Sem III
-- Target  : MySQL 8.0.16+ (CHECK constraints are enforced from this version)
-- Run in : MySQL Workbench (File > Open SQL Script > Execute)
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-- 1. CREATE THE DATABASE ------------------------------------------
DROP DATABASE IF EXISTS airline_reservation_system;
CREATE DATABASE airline_reservation_system
    CHARACTER SET utf8mb4
    COLLATE utf8mb4_unicode_ci;
USE airline_reservation_system;

-- 2. CREATE TABLES (parents first, children after) ----------------

-- 2.1 Passenger
CREATE TABLE Passenger (
    Passenger_ID   INT AUTO_INCREMENT PRIMARY KEY,
    First_Name     VARCHAR(50)  NOT NULL,
    Last_Name      VARCHAR(50)  NOT NULL,
    Email          VARCHAR(100) NOT NULL UNIQUE,
    Phone          VARCHAR(15)  NOT NULL,
    Date_of_Birth  DATE         NOT NULL,
    Gender         VARCHAR(10),
    CONSTRAINT chk_passenger_gender
        CHECK (Gender IN ('Male', 'Female', 'Other') OR Gender IS NULL)
) ENGINE=InnoDB;

-- 2.2 Airport
CREATE TABLE Airport (
    Airport_ID      INT AUTO_INCREMENT PRIMARY KEY,
    Airport_Code    VARCHAR(10)  NOT NULL UNIQUE,          -- e.g. BLR, DEL
    Airport_Name    VARCHAR(100) NOT NULL,
    City            VARCHAR(50)  NOT NULL,
    Country         VARCHAR(50)  NOT NULL,
    Terminal_Count  INT          NOT NULL DEFAULT 1,
    CONSTRAINT chk_airport_terminals CHECK (Terminal_Count > 0)
) ENGINE=InnoDB;

-- 2.3 Aircraft
CREATE TABLE Aircraft (
    Aircraft_ID      INT AUTO_INCREMENT PRIMARY KEY,
    Aircraft_Model   VARCHAR(50) NOT NULL,
    Registration_No  VARCHAR(30) NOT NULL UNIQUE,
    Total_Seats      INT         NOT NULL,
    Aircraft_Status  VARCHAR(20) NOT NULL DEFAULT 'Active',
    CONSTRAINT chk_aircraft_seats  CHECK (Total_Seats > 0),
    CONSTRAINT chk_aircraft_status CHECK (Aircraft_Status IN ('Active',
'Maintenance'))
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) ENGINE=InnoDB;

-- 2.4 Flight  (two foreign keys to Airport: departure and arrival)
CREATE TABLE Flight (
    Flight_ID             INT AUTO_INCREMENT PRIMARY KEY,
    Flight_Number         VARCHAR(20) NOT NULL,
    Aircraft_ID           INT         NOT NULL,
    Departure_Airport_ID  INT         NOT NULL,
    Arrival_Airport_ID    INT         NOT NULL,
    Departure_DateTime    DATETIME    NOT NULL,
    Arrival_DateTime      DATETIME    NOT NULL,
    Flight_Status         VARCHAR(20) NOT NULL DEFAULT 'Scheduled',
    CONSTRAINT fk_flight_aircraft
        FOREIGN KEY (Aircraft_ID) REFERENCES Aircraft(Aircraft_ID),
    CONSTRAINT fk_flight_departure
        FOREIGN KEY (Departure_Airport_ID) REFERENCES Airport(Airport_ID),
    CONSTRAINT fk_flight_arrival
        FOREIGN KEY (Arrival_Airport_ID) REFERENCES Airport(Airport_ID),
    CONSTRAINT uq_flight_number_departure
        UNIQUE (Flight_Number, Departure_DateTime),
    CONSTRAINT chk_flight_airports
        CHECK (Departure_Airport_ID <> Arrival_Airport_ID),
    CONSTRAINT chk_flight_times
        CHECK (Arrival_DateTime > Departure_DateTime),
    CONSTRAINT chk_flight_status
        CHECK (Flight_Status IN ('Scheduled', 'Delayed', 'Cancelled'))
) ENGINE=InnoDB;

-- 2.5 Seat
CREATE TABLE Seat (
    Seat_ID      INT AUTO_INCREMENT PRIMARY KEY,
    Aircraft_ID  INT         NOT NULL,
    Seat_Number  VARCHAR(10) NOT NULL,                    -- e.g. 12A
    Seat_Class   VARCHAR(20) NOT NULL DEFAULT 'Economy',
    Seat_Status  VARCHAR(20) NOT NULL DEFAULT 'Available',
    CONSTRAINT fk_seat_aircraft
        FOREIGN KEY (Aircraft_ID) REFERENCES Aircraft(Aircraft_ID),
    CONSTRAINT uq_seat_per_aircraft UNIQUE (Aircraft_ID, Seat_Number),
    CONSTRAINT chk_seat_class  CHECK (Seat_Class  IN ('Economy', 'Business',
'First')),
    CONSTRAINT chk_seat_status CHECK (Seat_Status IN ('Available',
'Reserved'))
) ENGINE=InnoDB;

-- 2.6 Booking  (associative entity: resolves Passenger M:N Flight)
CREATE TABLE Booking (
    Booking_ID      INT AUTO_INCREMENT PRIMARY KEY,
    Passenger_ID    INT           NOT NULL,
    Flight_ID       INT           NOT NULL,
    Seat_ID         INT           NOT NULL,
    Booking_Date    DATE          NOT NULL,
    Booking_Status  VARCHAR(20)   NOT NULL DEFAULT 'Pending',
    Fare            DECIMAL(10,2) NOT NULL,
    CONSTRAINT fk_booking_passenger
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        FOREIGN KEY (Passenger_ID) REFERENCES Passenger(Passenger_ID),
    CONSTRAINT fk_booking_flight
        FOREIGN KEY (Flight_ID) REFERENCES Flight(Flight_ID),
    CONSTRAINT fk_booking_seat
        FOREIGN KEY (Seat_ID) REFERENCES Seat(Seat_ID),
    -- a seat can be sold only once on a given flight (prevents double booking)
    CONSTRAINT uq_booking_flight_seat UNIQUE (Flight_ID, Seat_ID),
    CONSTRAINT chk_booking_status
        CHECK (Booking_Status IN ('Confirmed', 'Cancelled', 'Pending')),
    CONSTRAINT chk_booking_fare CHECK (Fare >= 0)
) ENGINE=InnoDB;

-- 2.7 Ticket  (1:1 with Booking)
CREATE TABLE Ticket (
    Ticket_ID      INT AUTO_INCREMENT PRIMARY KEY,
    Booking_ID     INT         NOT NULL UNIQUE,             -- UNIQUE enforces 1:1
    Ticket_Number  VARCHAR(30) NOT NULL UNIQUE,
    Issue_Date     DATE        NOT NULL,
    Ticket_Status  VARCHAR(20) NOT NULL DEFAULT 'Valid',
    CONSTRAINT fk_ticket_booking
        FOREIGN KEY (Booking_ID) REFERENCES Booking(Booking_ID),
    CONSTRAINT chk_ticket_status CHECK (Ticket_Status IN ('Valid', 'Cancelled'))
) ENGINE=InnoDB;

-- 2.8 Payment
CREATE TABLE Payment (
    Payment_ID      INT AUTO_INCREMENT PRIMARY KEY,
    Booking_ID      INT           NOT NULL,                 -- not UNIQUE: failed payments may be retried
    Payment_Date    DATE          NOT NULL,
    Payment_Method  VARCHAR(30)   NOT NULL,
    Amount          DECIMAL(10,2) NOT NULL,
    Payment_Status  VARCHAR(20)   NOT NULL DEFAULT 'Pending',
    Transaction_ID  VARCHAR(100)  NOT NULL UNIQUE,
    CONSTRAINT fk_payment_booking
        FOREIGN KEY (Booking_ID) REFERENCES Booking(Booking_ID),
    CONSTRAINT chk_payment_method
        CHECK (Payment_Method IN ('Card', 'UPI', 'Net Banking')),
    CONSTRAINT chk_payment_status
        CHECK (Payment_Status IN ('Successful', 'Failed', 'Pending')),
    CONSTRAINT chk_payment_amount CHECK (Amount > 0)
) ENGINE=InnoDB;

-- 2.9 Baggage  (1:1 with Booking)
CREATE TABLE Baggage (
    Baggage_ID          INT AUTO_INCREMENT PRIMARY KEY,
    Booking_ID          INT           NOT NULL UNIQUE,       -- UNIQUE enforces 1:1
    Baggage_Count       INT           NOT NULL DEFAULT 0,
    Total_Weight        DECIMAL(5,2)  NOT NULL DEFAULT 0,
    Excess_Baggage_Fee  DECIMAL(10,2) NOT NULL DEFAULT 0,
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    CONSTRAINT fk_baggage_booking
        FOREIGN KEY (Booking_ID) REFERENCES Booking(Booking_ID),
    CONSTRAINT chk_baggage_values
        CHECK (Baggage_Count >= 0 AND Total_Weight >= 0 AND Excess_Baggage_Fee >= 0)
) ENGINE=InnoDB;

-- 3. VERIFY THE IMPLEMENTATION ----------------------------------------
SHOW TABLES;

DESCRIBE Passenger;
DESCRIBE Airport;
DESCRIBE Aircraft;
DESCRIBE Flight;
DESCRIBE Seat;
DESCRIBE Booking;
DESCRIBE Ticket;
DESCRIBE Payment;
DESCRIBE Baggage;

-- List every foreign key relationship that was created
SELECT TABLE_NAME, COLUMN_NAME, CONSTRAINT_NAME,
       REFERENCED_TABLE_NAME, REFERENCED_COLUMN_NAME
FROM   INFORMATION_SCHEMA.KEY_COLUMN_USAGE
WHERE  TABLE_SCHEMA = 'airline_reservation_system'
  AND  REFERENCED_TABLE_NAME IS NOT NULL
ORDER  BY TABLE_NAME, CONSTRAINT_NAME;

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